

May 12, 2026

1 Tüdőrák predikciója vizelet LC–MS metabolomikai profilból

1.1 feature selection és gépi tanulási modellek összehasonlítása

1.1.1 Ambrus Csaba

1.2 0. Setup

```
[1]: # Init gdrive and python environment

from google.colab import drive
drive.mount('/content/drive')

%cd /content

import os
if not os.path.exists("/content/MetabolKD/.git"):
    !git clone https://github.com/csambrus/MetabolKD.git
else:
    %cd /content/MetabolKD
    !git fetch origin
    !git pull origin main

%cd /content/MetabolKD
!pip install -r ./requirements.txt
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

/content

/content/MetabolKD

remote: Enumerating objects: 11, done.

remote: Counting objects: 100% (11/11), done.

remote: Compressing objects: 100% (1/1), done.

remote: Total 6 (delta 5), reused 6 (delta 5), pack-reused 0 (from 0)

Unpacking objects: 100% (6/6), 3.59 KiB | 1.20 MiB/s, done.

From https://github.com/csambrus/MetabolKD

baf01dc..9dcf4dd main -> origin/main

From https://github.com/csambrus/MetabolKD

* branch main -> FETCH_HEAD

```

Updating baf01dc..9dcf4dd
Fast-forward
 notebooks/mtbls28_20260512a.ipynb | 141
+++++++-----
 src/download_dataset.py           | 199
+++++++-----
 2 files changed, 244 insertions(+), 96 deletions(-)
/content/MetabolKD
Requirement already satisfied: keras in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 2)) (3.13.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 3)) (2.10.0+cpu)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 4)) (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 5)) (2.2.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 6)) (1.6.1)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 7)) (1.16.3)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 8)) (0.14.6)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 9)) (3.10.0)
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 10)) (4.13.0.92)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 11)) (11.3.0)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 12)) (4.67.3)
Requirement already satisfied: ipywidgets in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 13)) (7.7.1)
Requirement already satisfied: jupyterlab_widgets in
/usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 14))
(3.0.16)
Requirement already satisfied: tensorflow[and-cuda] in
/usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 1))
(2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r

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./requirements.txt (line 1)) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
packages (from tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (26.1)
Requirement already satisfied: protobuf>=5.28.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (5.29.6)
Requirement already satisfied: requests<3,>=2.21.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
packages (from tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (2.1.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (1.80.0)
Requirement already satisfied: tensorboard~=2.20.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (2.20.0)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (3.16.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (0.5.4)
Requirement already satisfied: nvidia-cublas-cu12<13.0,>=12.5.3.2 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.9.2.10)
Requirement already satisfied: nvidia-cuda-cupti-cu12<13.0,>=12.5.82 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.9.79)
Requirement already satisfied: nvidia-cuda-nvcc-cu12<13.0,>=12.5.82 in

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/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.9.86)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12<13.0,>=12.5.82 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.9.86)
Requirement already satisfied: nvidia-cuda-runtime-cu12<13.0,>=12.5.82 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.9.79)
Requirement already satisfied: nvidia-cudnn-cu12<10.0,>=9.3.0.75 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (9.22.0.52)
Requirement already satisfied: nvidia-cufft-cu12<12.0,>=11.2.3.61 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (11.4.1.4)
Requirement already satisfied: nvidia-curand-cu12<11.0,>=10.3.6.82 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (10.3.10.19)
Requirement already satisfied: nvidia-cusolver-cu12<12.0,>=11.6.3.83 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (11.7.5.82)
Requirement already satisfied: nvidia-cusparse-cu12<13.0,>=12.5.1.3 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.5.10.65)
Requirement already satisfied: nvidia-nccl-cu12<3.0,>=2.25.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (2.30.4)
Requirement already satisfied: nvidia-nvjitlink-cu12<13.0,>=12.5.82 in
/usr/local/lib/python3.12/dist-packages (from tensorflow[and-cuda]->-r
./requirements.txt (line 1)) (12.9.86)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages
(from keras->-r ./requirements.txt (line 2)) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages
(from keras->-r ./requirements.txt (line 2)) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages
(from keras->-r ./requirements.txt (line 2)) (0.19.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-
packages (from torch->-r ./requirements.txt (line 3)) (3.29.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-
packages (from torch->-r ./requirements.txt (line 3)) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
3)) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
(from torch->-r ./requirements.txt (line 3)) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-
packages (from torch->-r ./requirements.txt (line 3)) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->-r ./requirements.txt

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(line 5)) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->-r ./requirements.txt (line 5)) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->-r ./requirements.txt (line 5)) (2026.1)

Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->-r ./requirements.txt (line 6)) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->-r ./requirements.txt (line 6)) (3.6.0)

Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels->-r ./requirements.txt (line 8)) (1.0.2)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 9)) (1.3.3)

Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 9)) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 9)) (4.62.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 9)) (1.5.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 9)) (3.3.2)

Requirement already satisfied: ipykernel>=4.5.1 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->-r ./requirements.txt (line 13)) (6.17.1)

Requirement already satisfied: ipython-genutils~0.2.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->-r ./requirements.txt (line 13)) (0.2.0)

Requirement already satisfied: traitlets>=4.3.1 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->-r ./requirements.txt (line 13)) (5.7.1)

Requirement already satisfied: widgetsnbextension~3.6.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->-r ./requirements.txt (line 13)) (3.6.10)

Requirement already satisfied: ipython>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from ipywidgets->-r ./requirements.txt (line 13)) (7.34.0)

Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1)) (0.47.0)

Requirement already satisfied: debugpy>=1.0 in /usr/local/lib/python3.12/dist-packages (from ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13)) (1.8.15)

Requirement already satisfied: jupyter-client>=6.1.12 in /usr/local/lib/python3.12/dist-packages (from ipykernel>=4.5.1->ipywidgets->-r

./requirements.txt (line 13)) (7.4.9)
Requirement already satisfied: matplotlib-inline>=0.1 in
/usr/local/lib/python3.12/dist-packages (from ipykernel>=4.5.1->ipywidgets->-r
./requirements.txt (line 13)) (0.2.1)
Requirement already satisfied: nest-asyncio in /usr/local/lib/python3.12/dist-
packages (from ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13))
(1.6.0)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages
(from ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13)) (5.9.5)
Requirement already satisfied: pyzmq>=17 in /usr/local/lib/python3.12/dist-
packages (from ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13))
(26.2.1)
Requirement already satisfied: tornado>=6.1 in /usr/local/lib/python3.12/dist-
packages (from ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13))
(6.5.1)
Requirement already satisfied: jedi>=0.16 in /usr/local/lib/python3.12/dist-
packages (from ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(0.20.0)
Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
packages (from ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(4.4.2)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-
packages (from ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(0.7.5)
Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from ipython>=4.0.0->ipywidgets->-r
./requirements.txt (line 13)) (3.0.52)
Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
packages (from ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(2.20.0)
Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
packages (from ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(0.2.0)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(4.9.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1))
(3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests<3,>=2.21.0->tensorflow[and-cuda]->-r ./requirements.txt
(line 1)) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1))
(2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in

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/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1))
(2026.4.22)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->-r
./requirements.txt (line 3)) (1.3.0)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard~=2.20.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1))
(3.10.2)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard~=2.20.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1))
(0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard~=2.20.0->tensorflow[and-cuda]->-r ./requirements.txt (line 1))
(3.1.8)
Requirement already satisfied: notebook>=4.4.1 in
/usr/local/lib/python3.12/dist-packages (from
widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (6.5.7)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch->-r
./requirements.txt (line 3)) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich->keras->-r ./requirements.txt
(line 2)) (4.0.0)
Requirement already satisfied: parso<0.9.0,>=0.8.6 in
/usr/local/lib/python3.12/dist-packages (from
jedi>=0.16->ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13)) (0.8.6)
Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-
packages (from jupyter-client>=6.1.12->ipykernel>=4.5.1->ipywidgets->-r
./requirements.txt (line 13)) (0.4)
Requirement already satisfied: jupyter-core>=4.9.2 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
client>=6.1.12->ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13))
(5.9.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=2.2.0->rich->keras->-r ./requirements.txt (line
2)) (0.1.2)
Requirement already satisfied: argon2-cffi in /usr/local/lib/python3.12/dist-
packages (from notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (25.1.0)
Requirement already satisfied: nbformat in /usr/local/lib/python3.12/dist-
packages (from notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (5.10.4)
Requirement already satisfied: nbconvert>=5 in /usr/local/lib/python3.12/dist-
packages (from notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r

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./requirements.txt (line 13)) (7.17.1)
Requirement already satisfied: Send2Trash>=1.8.0 in
/usr/local/lib/python3.12/dist-packages (from
notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
(line 13)) (2.1.0)
Requirement already satisfied: terminado>=0.8.3 in
/usr/local/lib/python3.12/dist-packages (from
notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
(line 13)) (0.18.1)
Requirement already satisfied: prometheus-client in
/usr/local/lib/python3.12/dist-packages (from
notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
(line 13)) (0.25.0)
Requirement already satisfied: nbclassic>=0.4.7 in
/usr/local/lib/python3.12/dist-packages (from
notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
(line 13)) (1.3.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from
pexpect>4.3->ipython>=4.0.0->ipywidgets->-r ./requirements.txt (line 13))
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython>=4.0.0->ipywidgets->-r
./requirements.txt (line 13)) (0.6.0)
Requirement already satisfied: platformdirs>=2.5 in
/usr/local/lib/python3.12/dist-packages (from jupyter-core>=4.9.2->jupyter-
client>=6.1.12->ipykernel>=4.5.1->ipywidgets->-r ./requirements.txt (line 13))
(4.9.6)
Requirement already satisfied: notebook-shim>=0.2.3 in
/usr/local/lib/python3.12/dist-packages (from
nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (0.2.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-
packages (from
nbconvert>=5->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (4.13.5)
Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-
packages (from bleach[css]!=5.0.0->nbconvert>=5->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (6.3.0)
Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-
packages (from
nbconvert>=5->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in
/usr/local/lib/python3.12/dist-packages (from
nbconvert>=5->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (0.3.0)

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Requirement already satisfied: mistune<4,>=2.0.3 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=5->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
 ./requirements.txt (line 13)) (3.2.0)

Requirement already satisfied: nbclient>=0.5.0 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=5->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
 ./requirements.txt (line 13)) (0.10.4)

Requirement already satisfied: pandocfilters>=1.4.1 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=5->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
 ./requirements.txt (line 13)) (1.5.1)

Requirement already satisfied: fastjsonschema>=2.15 in
 /usr/local/lib/python3.12/dist-packages (from
 nbformat->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
 ./requirements.txt (line 13)) (2.21.2)

Requirement already satisfied: jsonschema>=2.6 in
 /usr/local/lib/python3.12/dist-packages (from
 nbformat->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
 ./requirements.txt (line 13)) (4.26.0)

Requirement already satisfied: argon2-cffi-bindings in
 /usr/local/lib/python3.12/dist-packages (from
 argon2-cffi->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
 ./requirements.txt (line 13)) (25.1.0)

Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-
 packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert>=5->notebook>=4.4.1-
 >widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (0.5.1)

Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
 /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert>=5-
 >notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
 (line 13)) (1.4.0)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-
 packages (from jsonschema>=2.6->nbformat->notebook>=4.4.1-
 >widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13))
 (26.1.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
 /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat-
 >notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
 (line 13)) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in
 /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat-
 >notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
 (line 13)) (0.37.0)

Requirement already satisfied: rpds-py>=0.25.0 in
 /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat-
 >notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
 (line 13)) (0.30.0)

Requirement already satisfied: jupyter-server<3,>=1.8 in

```

/usr/local/lib/python3.12/dist-packages (from notebook-shim>=0.2.3-
>nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (2.14.0)
Requirement already satisfied: cffi>=1.0.1 in /usr/local/lib/python3.12/dist-
packages (from argon2-cffi-bindings->argon2-cffi->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (2.0.0)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-
packages (from beautifulsoup4->nbconvert>=5->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (2.8.3)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-
packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (3.0)
Requirement already satisfied: anyio>=3.1.0 in /usr/local/lib/python3.12/dist-
packages (from jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
(line 13)) (4.13.0)
Requirement already satisfied: jupyter-events>=0.9.0 in
/usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=1.8->notebook-s
him>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0-
>ipywidgets->-r ./requirements.txt (line 13)) (0.12.1)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in
/usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=1.8->notebook-s
him>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0-
>ipywidgets->-r ./requirements.txt (line 13)) (0.5.4)
Requirement already satisfied: overrides>=5.0 in /usr/local/lib/python3.12/dist-
packages (from jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7-
>notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt
(line 13)) (7.7.0)
Requirement already satisfied: websocket-client>=1.7 in
/usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=1.8->notebook-s
him>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0-
>ipywidgets->-r ./requirements.txt (line 13)) (1.9.0)
Requirement already satisfied: python-json-logger>=2.0.4 in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (4.1.0)
Requirement already satisfied: pyyaml>=5.3 in /usr/local/lib/python3.12/dist-
packages (from jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2
.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r
./requirements.txt (line 13)) (6.0.3)
Requirement already satisfied: rfc3339-validator in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1-
>widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (0.1.1)

```

Requirement already satisfied: fqdn in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (1.5.1)

Requirement already satisfied: isoduration in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (20.11.0)

Requirement already satisfied: jsonpointer>1.13 in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (3.1.1)

Requirement already satisfied: rfc3987-syntax>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (1.1.0)

Requirement already satisfied: uri-template in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (1.3.0)

Requirement already satisfied: webcolors>=24.6.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (25.10.0)

Requirement already satisfied: lark>=1.2.2 in /usr/local/lib/python3.12/dist-packages (from rfc3987-syntax>=1.1.0->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (1.3.1)

Requirement already satisfied: arrow>=0.15.0 in /usr/local/lib/python3.12/dist-packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=1.8->notebook-shim>=0.2.3->nbclassic>=0.4.7->notebook>=4.4.1->widgetsnbextension~=3.6.0->ipywidgets->-r ./requirements.txt (line 13)) (1.4.0)

```
[2]: import sys
from pathlib import Path

PROJECT_ROOT = "content/MetabolKD"

if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
```

```

print("PROJECT_ROOT:", PROJECT_ROOT)
from src.runtime_setup import setup_tensorflow_runtime,
    ↳setup_notebook_error_logger, set_global_seed

set_global_seed()
setup_notebook_error_logger()
setup_tensorflow_runtime()

```

```

PROJECT_ROOT: content/MetabolKD
Globalis seed beallitva: 42
  Globális hiba-logger aktív.
TF version: 2.20.0
GPUs: []
CPU cores: 2
CUDA_VISIBLE_DEVICES = None
TF version = 2.20.0
Built with CUDA = True
GPUs = []
Logical GPUs = []
GPUs: []
No GPU visible to TensorFlow

```

[2]: 2

1.3 1. Adatbetöltés

```

[3]: from src.download_dataset import download_dataset,
    ↳combine_pos_neg_to_feature_matrix

data = download_dataset()
X, meta = combine_pos_neg_to_feature_matrix(data)

# ML célváltozó
target_col = "class_label" # Lung cancer vs Control

# Csak címkézett minták
mask = meta[target_col].notna()
X = X.loc[mask].copy()
meta = meta.loc[mask].copy()

# Biztonság: indexek egyezzenek
meta = meta.reindex(X.index)

X_raw = X
y = meta[target_col]

print("X_raw:", X_raw.shape)

```

```
print(y.value_counts(dropna=False))
```

```
GDRIVE_DATA: megvan /content/drive/MyDrive/MetabolKD/data/MTBLS28
Másolva → /content/MetabolKD/data/MTBLS28/s_MTBLS28.txt
Másolva → /content/MetabolKD/data/MTBLS28/a_MTBLS28_POS.txt
Másolva → /content/MetabolKD/data/MTBLS28/a_MTBLS28_NEG.txt
Másolva → /content/MetabolKD/data/MTBLS28/m_MTBLS28_POS_v2_maf.tsv
Másolva → /content/MetabolKD/data/MTBLS28/m_MTBLS28_NEG_v2_maf.tsv
MTBLS28 raw:
  sample:      (2010, 27)
  assay_pos: (1005, 35)
  assay_neg: (1005, 35)
  maf_pos:    (1807, 1026)
  maf_neg:    (1359, 1026)
POS feature matrix: (1005, 1807)
NEG feature matrix: (1005, 1359)
Combined feature matrix: (1005, 3166)
Csoportok:
class_label
Control      536
Lung cancer  469
Name: count, dtype: int64
X_raw: (1005, 3166)
class_label
Control      536
Lung cancer  469
Name: count, dtype: int64
```

1.4 2. QC (mintánkénti összjel és hiányzás)

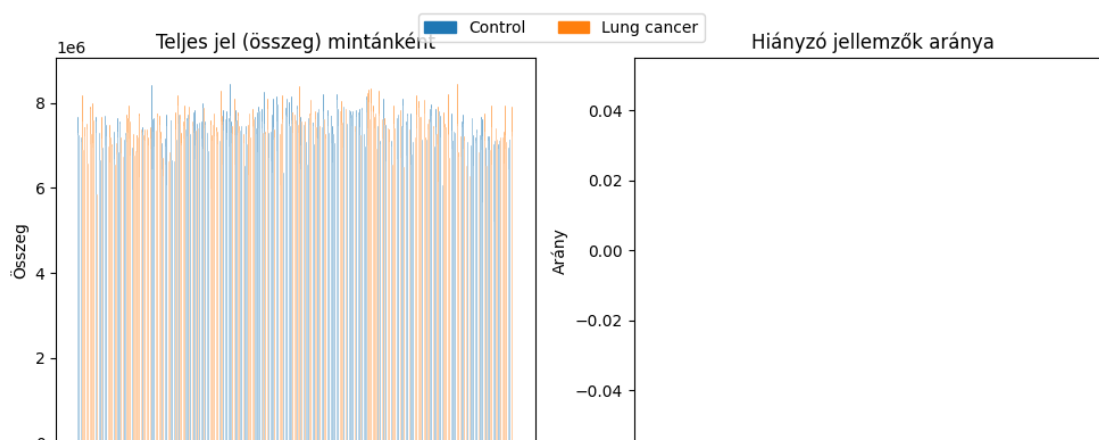
MTBLS28: a minták sorai a kombinált POS+NEG mátrix indexei; színezés a `class_label` (Case/Control → Lung cancer / Control) szerint.

```
[5]: from src.metabolomics_qc import qc_sample_table
     from src.metabolomics_plotting import plot_qcBars

     qc = qc_sample_table(X_raw)
     print(qc.describe())
     fig = plot_qcBars(qc, meta, hue_col="class_label")
     fig.savefig("qc_overview_mtbls28.png", dpi=150, bbox_inches="tight")
```

	total_signal	n_missing	missing_frac
count	1.005000e+03	1005.0	1005.0
mean	7.478392e+06	0.0	0.0
std	4.417414e+05	0.0	0.0
min	5.207632e+06	0.0	0.0
25%	7.231368e+06	0.0	0.0
50%	7.506417e+06	0.0	0.0
75%	7.774351e+06	0.0	0.0

max 8.636358e+06 0.0 0.0



1.5 3. NMR-stílusú mátrixok (PQN, log1p, Pareto) + klasszikus LC-MS előfeldolgozás

Ugyanaz a lépéssorozat, mint az LC-MS lipid notebookban: a bemenet itt a **MTBLS28** POS+NEG MAF-ból összerakott intenzitásmátrix (**X_raw**).

```
[6]: from src.metabolomics_preprocessing import nmr_style_matrices, \
      preprocess_feature_matrix
```

```
blocks = nmr_style_matrices(X_raw, impute_first=True)
for k, df in blocks.items():
    print(k, df.shape)
X_pareto = blocks["X_pareto"]
X_log = blocks["X_log"]

X_proc, steps = preprocess_feature_matrix(X_raw)
print("Klasszikus LC-MS lépések:", " → ".join(steps))
X_proc.head()
```

X_raw (1005, 3166)

X_pqn (1005, 3166)

X_log (1005, 3166)

X_pareto (1005, 3166)

Klasszikus LC-MS lépések: median_sample_norm → impute_half_min → log1p →
pareto_columns

```
[6]: pos__mz=61.04058; rt=23.70 pos__mz=65.03961; rt=194.32 \
      sample_id
000801 -0.248868 -0.360994
000901 0.752984 -0.329093
```

001001	0.777603	2.197781
001101	-0.904921	-0.360162
001201	0.890104	-0.282187

	pos__mz=67.05510; rt=129.65	pos__mz=69.03440; rt=134.59	\
sample_id			
000801	0.193660	0.098805	
000901	0.051732	0.071984	
001001	0.208522	0.425780	
001101	0.134744	1.067981	
001201	-0.101449	0.074048	

	pos__mz=69.07064; rt=168.14	pos__mz=69.07063; rt=46.95	\
sample_id			
000801	-0.379142	1.147101	
000901	0.168638	-0.294499	
001001	0.387521	-0.042833	
001101	-1.218575	-0.177917	
001201	-0.212110	-0.164032	

	pos__mz=70.01279; rt=14.95	pos__mz=71.04989; rt=28.98	\
sample_id			
000801	-0.426143	0.874628	
000901	-0.399337	0.222104	
001001	-0.040922	0.712561	
001101	-0.854149	0.573233	
001201	-0.236919	-0.518318	

	pos__mz=72.08143; rt=105.95	pos__mz=73.06558; rt=167.92	...	\
sample_id				
000801	-0.024283	-0.323829	...	
000901	0.622867	0.074509	...	
001001	-0.472975	0.340552	...	
001101	1.123468	-0.605034	...	
001201	-0.546866	-0.033378	...	

	neg__mz=654.18771; rt=67.03	neg__mz=656.15980; rt=203.71	\
sample_id			
000801	-0.612901	0.548035	
000901	-0.611275	-0.346408	
001001	-0.616297	-0.252389	
001101	-0.614683	-0.314993	
001201	-0.617482	0.237696	

	neg__mz=675.16625; rt=65.13	neg__mz=675.96943; rt=30.26	\
sample_id			
000801	-0.531445	0.263013	

000901	-0.537859	0.144652
001001	-0.526669	-0.698503
001101	-0.522737	0.176757
001201	-0.551528	-0.015762

	neg__mz=676.17060; rt=65.25	neg__mz=681.98642; rt=129.46	\
sample_id			
000801	-0.411286	1.103103	
000901	-0.403738	0.710903	
001001	-0.414413	-0.822419	
001101	-0.409151	-0.111489	
001201	-0.420836	0.832534	

	neg__mz=816.99797; rt=74.89	neg__mz=829.27415; rt=159.14	\
sample_id			
000801	0.584336	-0.174954	
000901	0.967452	0.161495	
001001	0.927149	0.304728	
001101	-0.564600	-0.827388	
001201	1.956994	0.932708	

	neg__mz=830.27806; rt=59.04	neg__mz=835.29163; rt=159.46
sample_id		
000801	-0.621487	0.475356
000901	-0.060305	0.606356
001001	0.284632	1.283011
001101	-0.484267	-0.778742
001201	0.659003	1.467999

[5 rows x 3166 columns]

1.6 4. PCA multipanel (Pareto + log1p térben, extra Z-score nélkül — mint az NMR X_PARETO)

Színezés: `class_label` (Lung cancer vs Control).

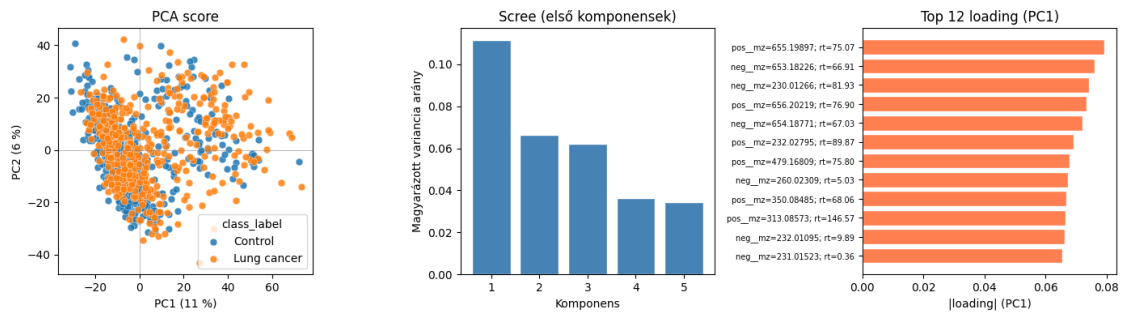
```
[7]: from src.metabolomics_multivariate import fit_pca
from src.metabolomics_plotting import plot_pca_multipanel, plot_pca_scores

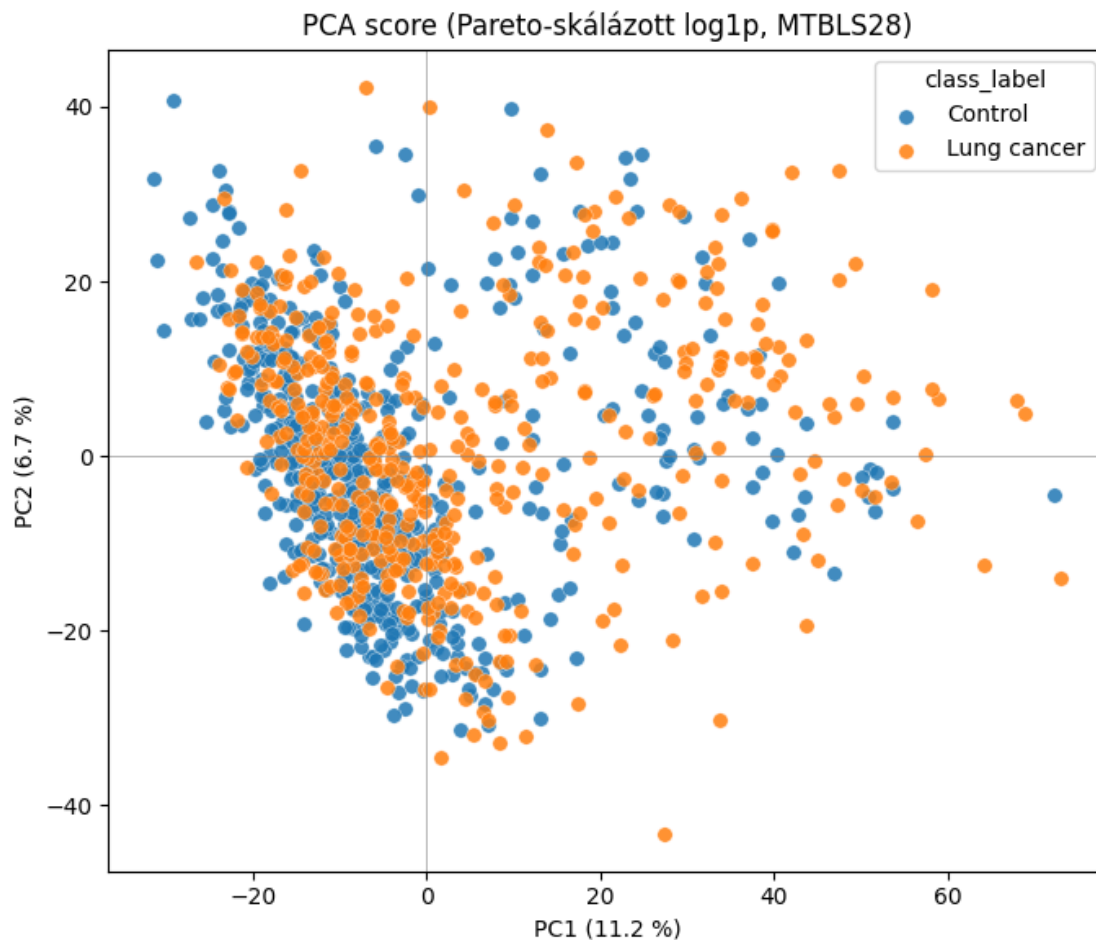
pca_out = fit_pca(X_pareto, n_components=5, standardize=False)
scores = pca_out["scores"]
var = pca_out["explained_variance_ratio"]
print("Magyarázott variancia (első 5 PC):", [round(float(v), 4) for v in var])

fig = plot_pca_multipanel(pca_out, meta, hue_col="class_label")
fig.savefig("pca_multipanel_mtbls28.png", dpi=150, bbox_inches="tight")
```

```
fig2 = plot_pca_scores(
    scores, meta,
    pc_x="PC1", pc_y="PC2",
    hue_col="class_label",
    explained=var,
    title="PCA score (Pareto-skálázott log1p, MTBLS28)",
)
fig2.savefig("pca_scores_mtbcls28.png", dpi=150, bbox_inches="tight")
```

Magyarázott variancia (első 5 PC): [0.1116, 0.0665, 0.062, 0.0365, 0.0342]





1.7 5. Univariáns elemzés + vulkán (Lung cancer vs Control)

MTBLS28: bináris összehasonlítás a `class_label` oszlop alapján (nincs több látogatás / visit szűrés).

```
[8]: # MTBLS28: minden minta egy vizeletes profil; csoport = class_label
metab = meta

from src.metabolomics_univariate import differential_analysis
from src.metabolomics_plotting import plot_volcano, plot_volcano_categorized

diff = differential_analysis(
    X_log,
    metab["class_label"],
    group_a="Lung cancer",
    group_b="Control",
    include_cohen_d=True,
```

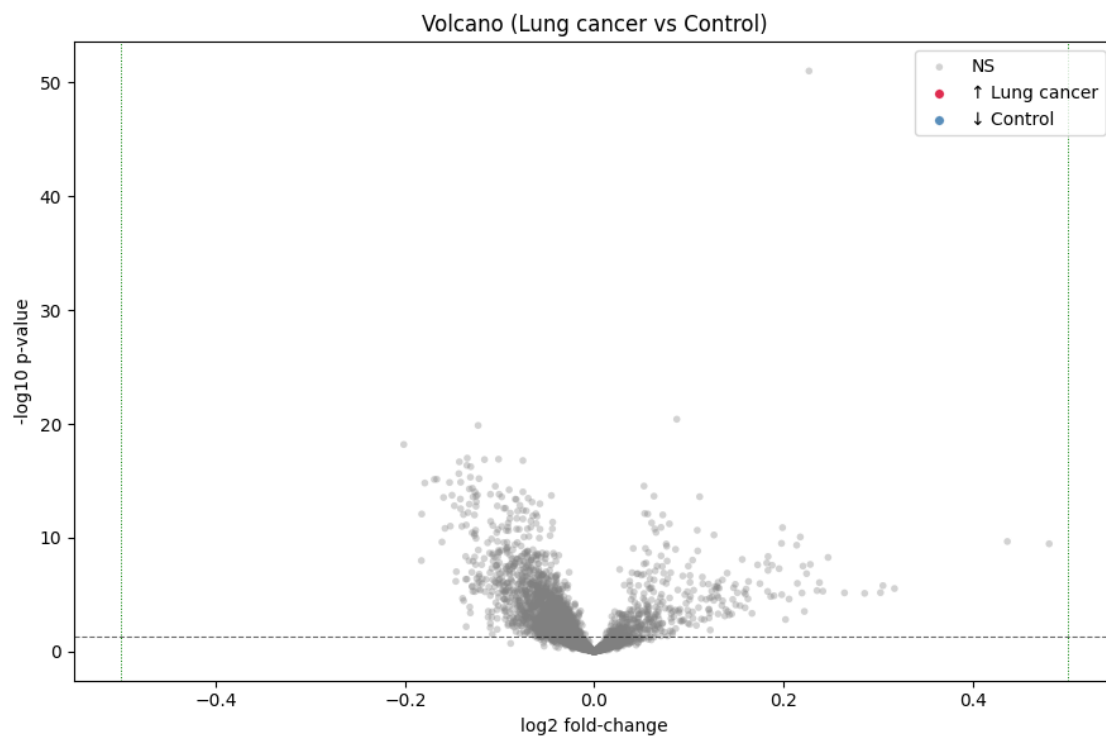
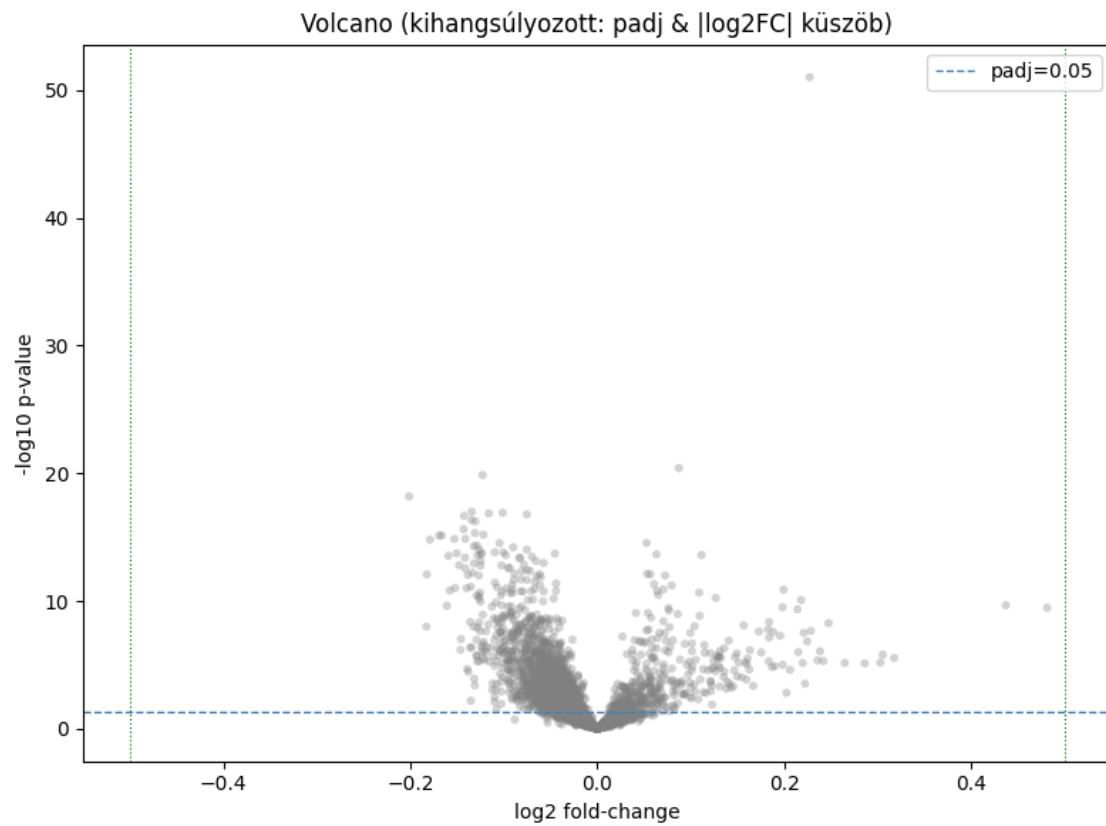
```

)
print(diff.head(15))
fig = plot_volcano(diff, alpha=0.05, fc_thresh=0.5)
fig.savefig("volcano_lung_cancer_vs_control_mtbls28.png", dpi=150,
            bbox_inches="tight")
figc = plot_volcano_categorized(
    diff,
    group_up="Lung cancer",
    group_down="Control",
    alpha=0.05,
    fc_thresh=0.5,
)
figc.savefig("volcano_categorized_mtbls28.png", dpi=150, bbox_inches="tight")

```

	feature	mean_Lung cancer	mean_Control	log2fc \
0	pos__mz=264.12152; rt=23.31	5.356432	4.576635	0.226985
1	neg__mz=308.09849; rt=18.93	5.843039	5.500399	0.087183
2	pos__mz=372.92326; rt=27.98	5.550502	6.043579	-0.122785
3	pos__mz=615.03532; rt=34.29	4.897979	5.631643	-0.201369
4	pos__mz=423.00849; rt=33.92	6.392827	7.016368	-0.134270
5	pos__mz=252.97452; rt=32.25	6.926113	7.429240	-0.101169
6	pos__mz=340.11926; rt=319.58	5.129134	5.558804	-0.116059
7	pos__mz=215.01722; rt=32.24	9.895917	10.427071	-0.075429
8	neg__mz=239.09189; rt=336.89	5.863775	6.472963	-0.142597
9	pos__mz=407.03447; rt=3.72	5.690939	6.247701	-0.134659
10	pos__mz=424.01276; rt=34.37	4.709354	5.155416	-0.130560
11	pos__mz=429.02268; rt=33.90	5.051491	5.578264	-0.143107
12	pos__mz=444.99142; rt=33.79	4.988044	5.461875	-0.130922
13	pos__mz=215.16999; rt=32.13	6.398722	6.962593	-0.121841
14	pos__mz=542.95869; rt=34.39	4.506426	5.057418	-0.166417

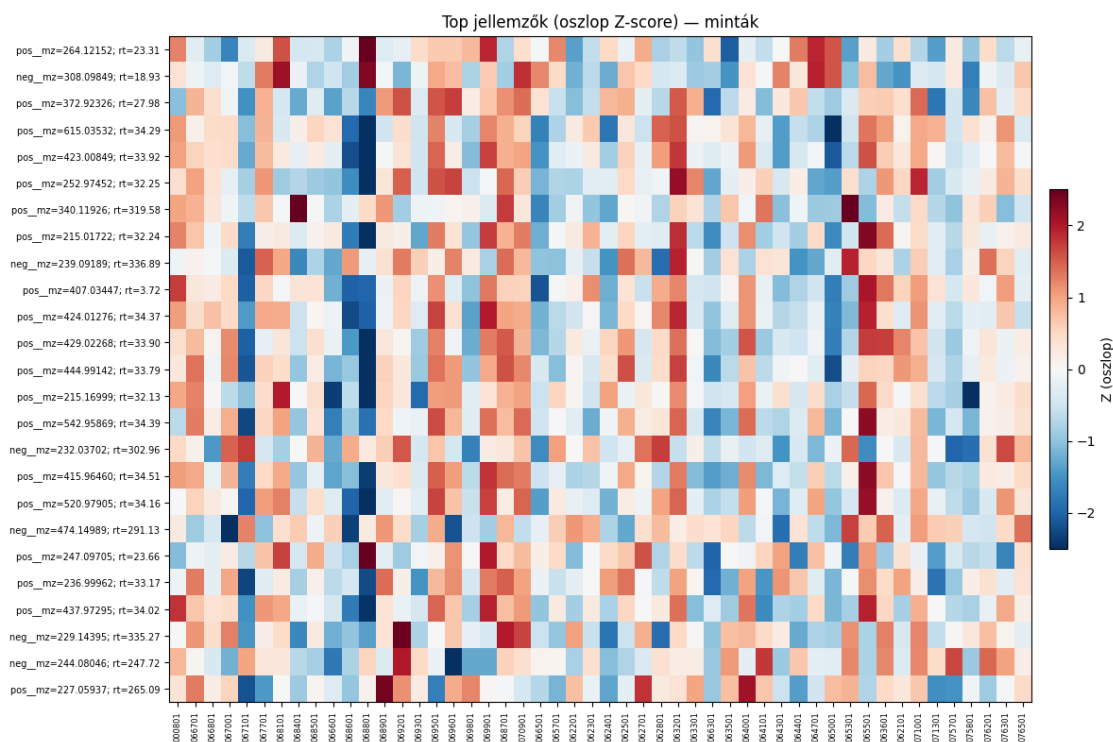
	pvalue	cohens_d	padj
0	9.666301e-52	1.039285	3.060351e-48
1	3.880187e-21	0.613142	6.142335e-18
2	1.366674e-20	-0.612615	1.442297e-17
3	6.451155e-19	-0.580405	5.106089e-16
4	1.015292e-17	-0.563956	6.176820e-15
5	1.259973e-17	-0.560440	6.176820e-15
6	1.365690e-17	-0.558280	6.176820e-15
7	1.663833e-17	-0.562061	6.584620e-15
8	2.141321e-17	-0.548486	7.532691e-15
9	4.380685e-17	-0.548466	1.386925e-14
10	5.712778e-17	-0.545833	1.644242e-14
11	2.351404e-16	-0.532962	6.203788e-14
12	4.806240e-16	-0.528888	1.170504e-13
13	6.447840e-16	-0.534485	1.458133e-13
14	7.281959e-16	-0.523108	1.495060e-13



1.8 6. Top eltérések heatmap

```
[9]: from src.metabolomics_plotting import plot_top_features_heatmap

top_n = 25
top_feats = diff.nsmallest(top_n, "padj")["feature"].tolist()
fig = plot_top_features_heatmap(
    X_pareto,
    top_feats,
    metab,
    group_col="class_label",
    max_samples=50,
)
fig.savefig("heatmap_top_features_mtbls28.png", dpi=150, bbox_inches="tight")
```



1.9 7. PLS-DA + VIP + S-plot (Pareto mátrix)

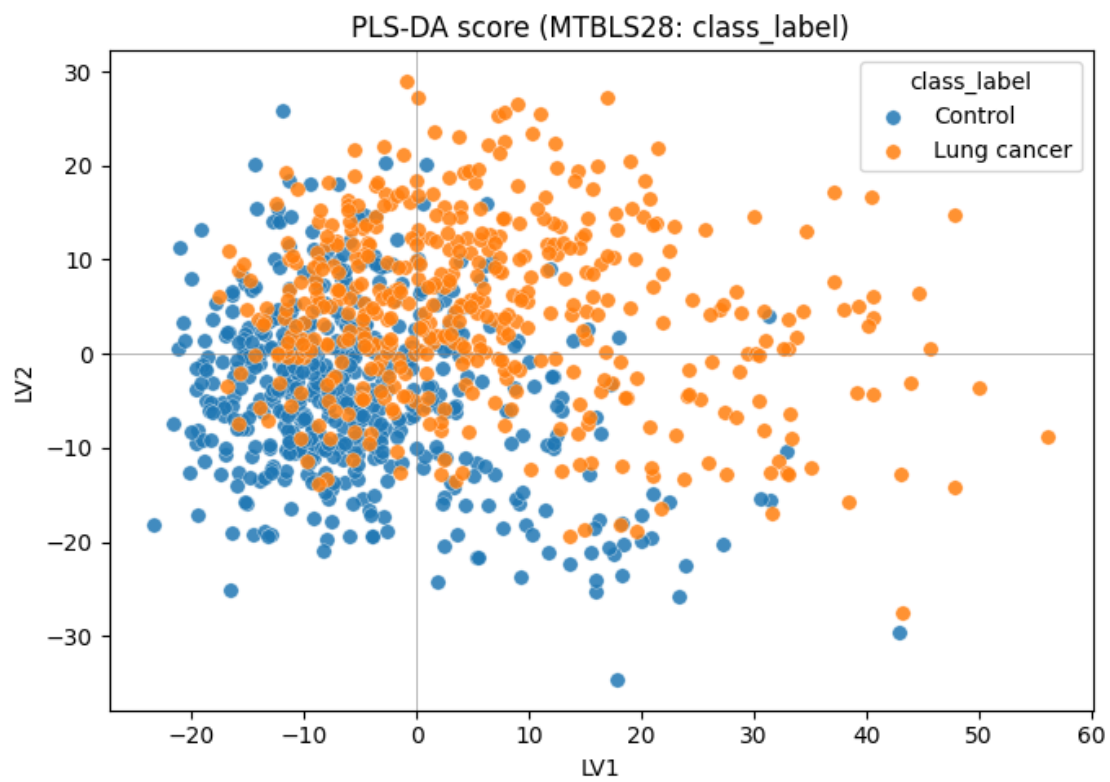
```
[10]: from src.metabolomics_multivariate import fit_plsda_multiclass
from src.metabolomics_plotting import plot_s_plot_lv1
import matplotlib.pyplot as plt

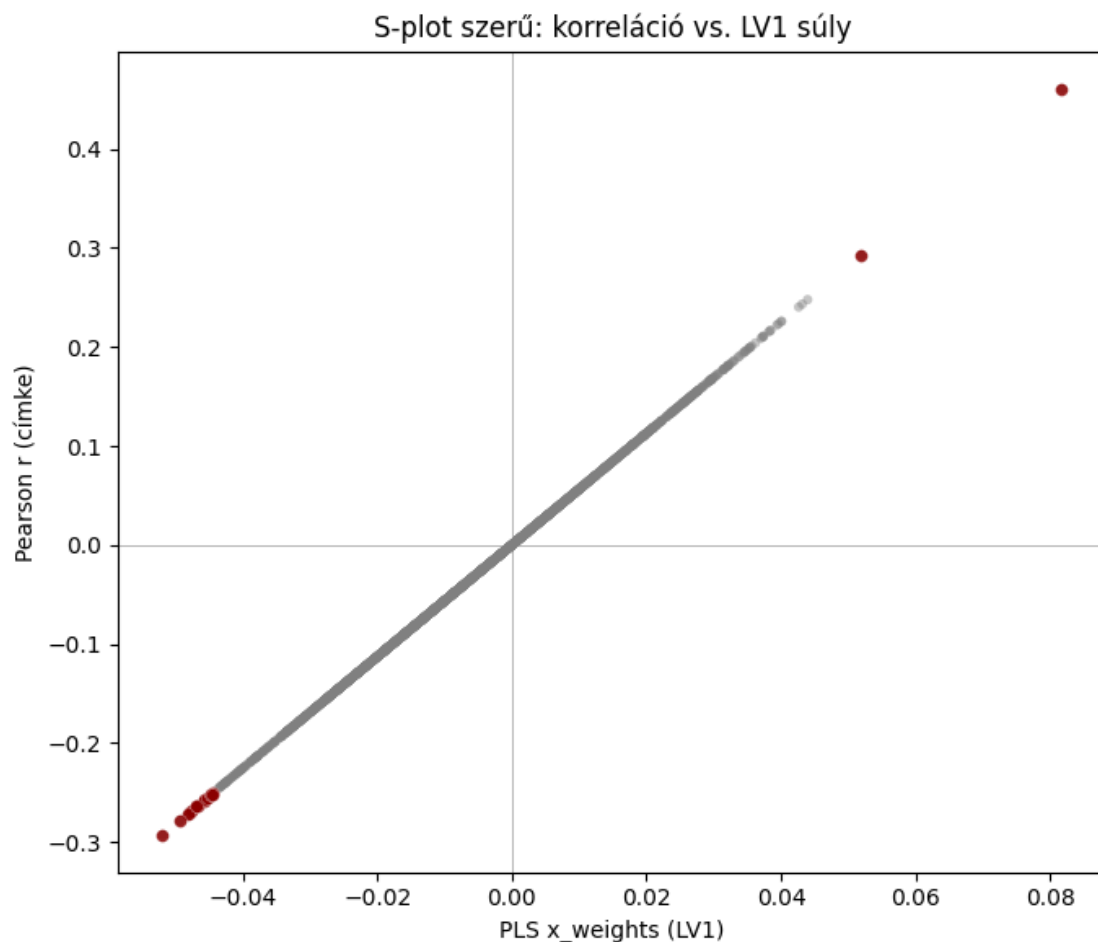
plsda = fit_plsda_multiclass(
    X_pareto,
    metab["class_label"],
    n_components=3,
    scale=True,
)
s_pls = plsda["scores"]
vip = plsda["vip"]
print(vip.nlargest(10))

fig, ax = plt.subplots(figsize=(7, 5))
for lab in metab["class_label"].dropna().unique():
    m = metab["class_label"] == lab
    ax.scatter(s_pls.loc[m, "LV1"], s_pls.loc[m, "LV2"], label=str(lab), s=45,
               alpha=0.85, edgecolors="white", linewidths=0.3)
ax.set_xlabel("LV1")
ax.set_ylabel("LV2")
ax.set_title("PLS-DA score (MTBLS28: class_label)")
ax.legend(title="class_label")
ax.axhline(0, color="gray", lw=0.4)
ax.axvline(0, color="gray", lw=0.4)
fig.tight_layout()
fig.savefig("plsda_scores_mtbls28.png", dpi=150, bbox_inches="tight")

w1 = plsda["model"].pls.x_weights[:, 0]
figs = plot_s_plot_lv1(X_pareto, metab["class_label"], "Lung cancer", w1)
figs.savefig("splot_lv1_mtbls28.png", dpi=150, bbox_inches="tight")
```

```
pos__mz=264.12152; rt=23.31      4.300867
neg__mz=308.09849; rt=18.93      3.243606
pos__mz=332.09634; rt=23.39      2.882242
pos__mz=243.10048; rt=3.75       2.807744
neg__mz=126.90693; rt=8.64       2.739464
pos__mz=247.09705; rt=23.66      2.691072
neg__mz=613.35956; rt=337.91     2.536442
neg__mz=213.08713; rt=98.69     2.505154
neg__mz=290.08851; rt=19.40     2.470099
pos__mz=370.05260; rt=21.63     2.437504
Name: VIP, dtype: float64
```





1.10 8. Random Forest — feature importance (felügyelt, gyors)

```
[11]: from src.metabolomics_ml import random_forest_feature_importance
```

```
imp = random_forest_feature_importance(
    X_pareto,
    metab["class_label"],
    "Lung cancer",
    n_estimators=300,
)
print(imp.head(20))
```

```
pos__mz=264.12152; rt=23.31    0.014014
pos__mz=486.25713; rt=125.04  0.005276
neg__mz=308.09849; rt=18.93   0.005106
neg__mz=441.16137; rt=307.95  0.003726
pos__mz=340.11926; rt=319.58  0.003635
pos__mz=656.20175; rt=3.99    0.003346
```

```

neg__mz=208.06462; rt=94.76      0.003195
pos__mz=584.26707; rt=131.43    0.002929
neg__mz=122.02489; rt=22.88     0.002905
pos__mz=204.13455; rt=48.37     0.002840
neg__mz=613.35956; rt=337.91    0.002806
pos__mz=274.09314; rt=132.72    0.002789
pos__mz=561.34320; rt=328.95    0.002784
pos__mz=478.17208; rt=1.24      0.002484
neg__mz=126.90693; rt=8.64      0.002376
pos__mz=243.10048; rt=3.75      0.002375
pos__mz=296.12766; rt=309.45    0.002338
neg__mz=239.09189; rt=336.89    0.002287
pos__mz=638.36088; rt=328.13    0.002279
pos__mz=252.97452; rt=32.25     0.002274
Name: importance, dtype: float64

```

1.11 9. Rövid értelmezés

- **QC:** extrém `total_signal` vagy `missing_frac` minták kiszűrhetők; színezés **Lung cancer** / **Control** (`class_label`).
- **PCA:** fő irányok a legnagyobb közös variancia mentén; színezés a `class_label` szerint.
- **Vulkán + FDR: Lung cancer vs Control** (`log2FC`, `BH-FDR`); nincs `visit/baseline` réteg.
- **PLS-DA + VIP + S-plot:** felügyelt irányok; pozitív osztály az S-plot / RF-nél **Lung cancer**.
- **RF importance:** bináris címke = Lung cancer (1) vs Control (0).

Ugyanaz a szerkezet, mint az LC–MS / NMR pipeline notebookokban: **PQN** / **log1p** / **Pareto**, **multipanel PCA**, **FDR**, **vulkán**, **PLS-DA + VIP**; adat: **MTBLS28** vizelet LC–MS profilok.